Opera Numerorum

After Euler, Riemann, Dirichlet

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Series Structure

Series	Title	Scope
Opera Numerorum I	Millennial Mathematics	BSD conjecture, Hodge conjecture, Riemann Hypothesis via Bost-Connes criterion. Mo
Opera Numerorum II	Morning Star Engineering	Phase-Z metric, PLL cascade, wormhole architecture, FTL transmission protocol, oper

Preface

This series began with a single question: can a machine certify a mathematical proof the way a notary certifies a document? Not approximately, not informally, but with a SHA-256 fingerprint that makes forgery computationally impossible. The answer, as the modules in this series demonstrate, is yes.

I am an independent researcher. I do not hold a PhD. I work in the tradition of those who approach mathematics because the questions will not let them go, not because an institution has commissioned the work. The history of mathematics has always had room for that kind of person. I hope it still does.

Opera Numerorum (the Works of Numbers) is the public name for what began as an internal working document. The internal codename, Battle Plan v1.6, is preserved in all SHA-bound certificate files to maintain causal chain integrity. Every numerical result in this series was computed in this environment, verified to match, and bound by its SHA-256 hash. No value was fabricated. No error was hidden.

The Mathematical Programme

The central object is the exceptional set S(pi/10): the set of primes p for which the fractional part of p * pi/10 is unusually small -- smaller than 1/p. Analogy: throw a dart at a number line; most darts land in random positions. An exceptional prime is one whose dart lands almost exactly on an integer. The question is how many such primes exist, and what their structure implies about the deep zeros of L-functions.

The Riemann Hypothesis is the statement that all non-trivial zeros of the Riemann zeta function lie on the line Re(s) = 1/2. This paper does not prove RH outright. It certifies a necessary component: GRH for a specific Hasse-Weil L-function associated to the modular curve X_0(143). The path from that result to ζ(s) runs through the arithmetic of the field Q(sqrt(-143)) and the theory of complex multiplication. The road is long but the first milestone is machine-certified.

The Bost-Connes criterion converts a combinatorial fact -- the finiteness of S(pi/10) -- into an analytic statement about L-functions. It does this through an energy function C(S), a sum of terms log(p)/(p-1) over the exceptional primes. When C(S) exceeds a threshold determined by the genus of the modular curve, GRH follows. This is, in some sense, the bridge between elementary number theory and the deepest open problem in mathematics.

The Machine Verification Protocol

Every claim in this series follows a four-step protocol. First, the computation is implemented in Python (mpmath at 64 decimal places) or C (80-bit long double for the kappa bound). Second, the program is run and its standard output is captured to a file. Third, the SHA-256 hash of that output file is computed and recorded. Fourth, each module cites the SHA of all upstream modules it depends on. Changing any upstream value breaks the chain. This is not peer review in the traditional sense -- it is something stronger: cryptographic reproducibility.

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The master manifest SHA (SHA256 of cat m1.out through m6.out) is:
5b80b84d1d3d13e216eeecd8155c1edc854d578e7d2dae9c4bc72fcbf7ebe3c9. Any reader may reproduce this
value by running: bash verify_all.sh
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A Note on Series II: Morning Star

The second series was not planned. It grew from the mathematics. The Phase-Z metric that describes wormhole geometry is structurally identical to the zero condition Z(r) -> 0 that appears in the exceptional prime analysis. The junction was unexpected, and I do not fully understand it yet. What I can say is that the computations are internally consistent, the numbers agree with each other across modules, and the SHA chain is unbroken.

I chose to write in the voice of Euler not to claim his authority but because his way of working -- careful, computational, unembarrassed by the concrete -- felt right for what this series is doing. He spent years computing specific values before the general theory came. This series is in that tradition.

> *I am a man of Basel, a man of numbers, but I am not blind. Every equation needs an origin. Every graph needs an axis. If we are to move between stars, let the zero of our clock be the moment He chose to give all. -- L.E. (Euler's Personal Log, L2 Station Morning Star, 2026)*

Acknowledgements

This work was carried out entirely in the Replit environment. I am grateful to the agent who served as computational partner throughout this series: patient, precise, and willing to run the binary search a fifth time when the numbers did not agree. Every error in this series was caught not hidden, and that discipline belongs to both of us.

I draw on three traditions: the Eulerian tradition of concrete computation as a path to general truth; the Islamic tradition of precision in the preservation of knowledge (it was the scribes of the East who gave us the numerals this paper uses); and the Christian tradition of Mercy as the ground of any serious act. These are not contradictions. They are the same circle, drawn by different hands.

I am an independent researcher actively seeking academic sponsorship and partnership. This work is freely shared with any staff member, shareholder, or member of the general public. I ask only that errors be reported to me and that the SHA chain be preserved intact in any reproduction.

Author Record

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ORCID will be inserted before final assembly. Author: David Fox (D.J.F.) -- Independent
researcher -- May 2026
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How to Read This Series

Each module in Opera Numerorum is self-contained. You can read Module 5 without reading Module 3, as long as you are willing to accept Module 3's SHA as a black box. If you want to verify the whole chain, run verify_all.sh. If you want to read the mathematics, start with Module 1 (alpha_0 = 299 + pi/10) and follow the dependency arrows.

Readers interested in the Millennial Mathematics results (BSD, Hodge, RH) should follow Series I: Modules M1 through M8. Readers interested in the engineering applications (Phase-Z thrust, wormhole architecture, Morning Star operations) should follow Series II: Modules M8C through M8M. The two series share a causal spine -- the alpha_0 constant computed in M1 appears in both.

A note on submission: arXiv does not require a PhD. It requires an endorsement from a current arXiv author in the relevant subject area (math.NT for number theory, hep-th for the physics modules). If you are reading this and you are in a position to endorse, I would welcome the conversation.
